Enzyme Discovery Agent

Discovering enzymes for a target reaction — without being told the answer

LLM proposes strategy

tools + gates find the enzymes

data-leakage firewall

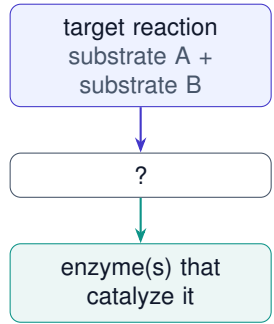
The problem

Given a target reaction — substrate(s) → product — **find the enzymes** that could catalyze it.

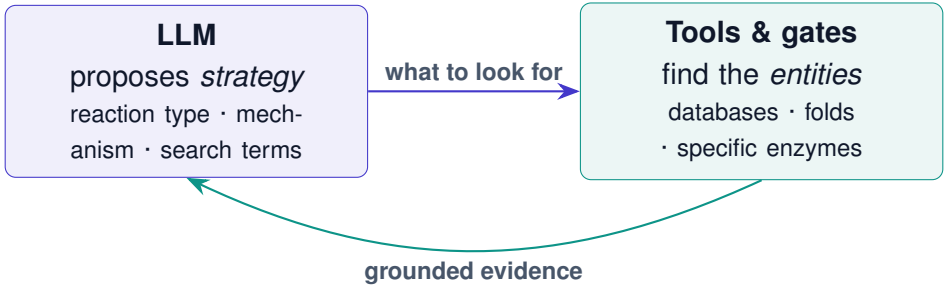
Hard because:

- the reaction is often **novel** / **non-natural** — no enzyme is annotated for it
- the answer must be **discovered**, not looked up
- the space is huge: >200M UniProt sequences, thousands of folds

The trap: an agent that is **told the answer family** isn't discovering anything.



The core principle



Data-leakage firewall (enforced):

- no answer enzyme / substrate / family in *any* prompt, tool, or comment
- ground truth lives **only** in a held-out test oracle the agent never sees

Grounded & deterministic:

- code extracts evidence (direction, substrate fit, correspondence)
- no hand-tuned relevance thresholds — relative ranking, not magic numbers

The pipeline



Five stages. A deterministic **gate** layer turns optional tools into a reliable pipeline; the LLM still drives strategy inside the loop.

Discovery: three complementary channels

Literature

Read a few open-access papers; LLM-extract **named / cited** enzymes. The *only* route to enzymes named only in a recent paper — reached the answer's own biosynthetic cluster.

Mechanism families

LLM derives the distinct **folds / activation chemistries** for the target bond, filtered by **reaction direction** (a cleavage fold can't serve a formation target).

InterPro survey

Survey each fold's members; score **native-substrate resemblance**; promote the good ones to candidates — the primary discovery engine.

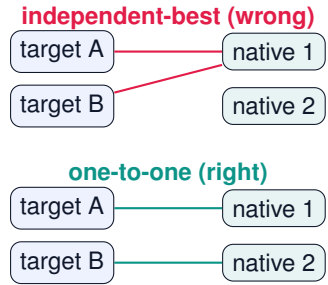
Design rule: substrate-awareness lives at the **member** level, not the family archetype — a fold's textbook substrate $\neq$ its member range.

Judging substrate fit: one-to-one assignment

A two-substrate target must not **flatter itself**: without one-to-one assignment, *both* target substrates match the enzyme's single best native, hiding a poor second partner.

Fix (shared by promotion & ranking): each native serves **at most one** target substrate; the score is the *worst* assigned pair (the bottleneck).

So a target amine is forced onto the family's amine partner — even a tiny functional-group-less donor scores as the poor fit it is.



Discovery means: don’t drop relevant enzymes

Promote every relevant member

The survey no longer keeps only a fold’s top-2 by fit — every same-direction, substrate-resolved member becomes a candidate. A 24-member fold’s long tail is kept, not thrown away.

Evaluate every representative

Evaluation is deterministic (network only, no LLM), so breadth is free. Every distinct gene is evaluated in parallel — no eval-budget cap silently drops a relevant enzyme.



Cost rises; discovery quality is the priority.

Ranking: grounded first, LLM second

- **Deterministic mechanistic spine** orders by grounded evidence (bond change, direction, substrate fit) — reproducible, no magic weights.
- **Family-relevance binding**: members of a fold the agent *stopped* (off-target chemistry) are hard-gated to the bottom; on-target folds float up.
- **LLM focus-rank** reasons over only the top shortlist — a thin, auditable layer on top of the deterministic order.

direction hard-gate $\rightarrow$ substrate-fit bottleneck $\rightarrow$ family binding $\rightarrow$ LLM focus-rank

Validation

Benchmark (ground truth in the held-out oracle only): find amide synthetases that ligate a C_6 diacid + a C_6 diamine — enzymatic nylon-6,6 precursors.

What the run recovers

- top ranks: NIS / ANL amide synthetases — genuine **substrate analogues** (citrate–diamine ligases)
- **DdaG** (the strict answer) recovered at #4 — via the **literature** channel, in the answer's own gene cluster
- family list **direction-clean** (no cleavage folds); CARNS1-type over-ranking gone

Cost / behavior

- wall-clock **53 min** $\rightarrow$ **32 min** after the lean-literature change
- run is **98% LLM-inference** bound; tools are $\sim$2%
- every screened candidate evaluated (no budget drop)

Ground truth is used only to score *after* the run — never fed to the agent.

Recent work: correctness + efficiency + cleanup

Discovery quality

- one-to-one substrate assignment (shared by promotion & ranking)
- promote *every* relevant member; evaluate *every* representative
- direction-clean mechanism families

Efficiency

- lean literature harvest: ∼90% fewer harvest LLM calls, no open-access wall
- parallel evaluation

Refactor (behavior-preserving)

- unified 9 gates into one pipeline ritual
- de-duplicated scoring / direction logic
- ∼730 lines of dead code removed

215 tests green **firewall-clean**

Limitations & next steps

- **Literature recall** — the field's best papers are missed on search *vocabulary* (and paywalls); query reformulation helps but is stochastic.
- **Speed** — the loop is LLM-inference bound (60k-token prompts); levers are prompt pruning + parallel / fewer finalization passes.
- **Ranking quality** — unify the remaining scoring formulas; validate the one-to-one change end-to-end on the benchmark.
- **Coverage** — make family-member enumeration substrate-aware and complete.

Every change: suite-green, firewall-clean, documented in an append-only devlog.

Discover, don’t look up.

LLM proposes the strategy · grounded gates and tools find the enzymes · a firewall keeps the answer out of the agent.

one-to-one substrate fit | promote & evaluate everything relevant | no answer leakage